Project Requirements Document: Cyclistic Station Dashboard

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Client/Sponsor: Jamal Harris, Director, Customer Data

Purpose: Stakeholders want to understand how their customers are using their bikes; their top priority is identifying customer demand at different station locations.Based on these insights they want a new tool (dashboard) that helps them monitor bike users and bike station usage. The future dashboard will be used to inform current usage and improve the platform for better users' experience

Key dependencies:

This project will require a dataset of customer data, so the Director of Customer Data will need to approve the request. Approval should also be given by the teams that own specific product data including bike trip duration and bike identification numbers to validate that the data is being interpreted correctly. Primary contacts are Adhira, Megan, Rick, and Tessa

Our team members are:

1. Adhira Patel, API Strategist
2. Megan Pirato, Data Warehousing Specialist
3. Rick Andersson, Manager, Data Governance
4. Tessa Blackwell, Data Analyst
5. Brianne Sand, Director, IT
6. Shareefah Hakimi, Project Manager

Deliverables:

- Understand how the current line of bikes are used and how new stations might alleviate demand in different geographical areas.

- How can we apply customer usage insights to inform new station growth?
- The customer growth team wants to understand how different users (subscribers and non-subscribers) use our bikes. We'll want to investigate a large group of users to get a fair representation of users across locations and with low- to high-activity levels.

Stakeholder requirements:

The future dashboard should contain the following metrics:

- A table or map visualization exploring starting and ending station locations, aggregated by location. I can use any location identifier, such as station, zip code, neighborhood, and/or borough. This should show the number of trips at starting locations - R
- A visualization showing which destination (ending) locations are popular based on the total trip minutes - R
- A visualization that focuses on trends from the summer of 2015 - D
- A visualization showing the percent growth in the number of trips year over year - R
- Gather insights about congestion at stations - N
- Gather insights about the number of trips across all starting and ending locations - R
- Gather insights about peak usage by time of day, season, and the impact of weather - R

Success criteria:

In 6 weeks creating a dashboard that includes a table or map visualization exploring starting and ending station locations, a visualization showing which destination (ending) locations are popular based on the total trip minutes, a visualization that focuses on trends from the summer of 2015 and a visualization showing the percent growth in the number of trips year over year.

Dashboard will be fully functional for the team to view insights into user behavior

Specific: The dashboard should help to understand the Cyclistic riders' current bike usage and the level of demand at different bike stations.

Measurable: During the work we need to assess the impact of different factors at trip counts. Does the number of trips change due to the weather conditions (high or low temperature, wind speed or the amount of precipitation)? Are there the most popular stations and where they are located?

Action-oriented: Based on the insights we get, the Cyclistic team will determine key aspects of product developments: creating new stations, adding additional bikes to the most popular locations (if they are), planning maintenance work when the demand is lower than average and so on.

Relevant: All metrics must support the primary question: How can we build a better Cyclistic experience?

Time-bound: We need to analyze at least one year of trips from our database to understand how season and weather factors influence bike usage.

User journeys:

The main purpose of Cyclistic is to provide customers with a better bike-share experience. Insights Gained insights will help the Cyclistic Team to develop the product that is more suitable for the customers' needs.

Assumptions:

The dataset includes latitude and longitude of stations but does not identify more geographic aggregation details like zip code, neighborhood name, or borough. The team will provide a separate database with this data.

The weather data provided does not include what time precipitation occurred; it's possible that on some days, it precipitated during off-peak hours. However, for the purpose of this dashboard, you should assume any amount of precipitation that occurred on the day of the trip could have an impact.

Starting bike trips at a location will be impossible if there are no bikes available at a station, so we might need to consider other factors for demand.

Compliance and privacy:

The data must not include any personal info (name, email, phone, address). Personal info is not necessary for this project

Accessibility:

Dashboard needs to be accessible, with large print and intuitively understandable.

People with dashboard-viewing privileges: Adhira, Brianne, Ernest, Jamal, Megan, Nina, Rick, Shareefah, Sara, Tessa

Roll-out plan:

- Week 1: Dataset assigned. Initial design for fields and BikeIDs validated to fit the requirements.
- Weeks 2–3: SQL and ETL development
- Weeks 3–4: Finalize SQL. Dashboard design. 1st draft review with peers.
- Weeks 5–6: Dashboard development and testing